



INTERNSHIP REPORT

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Table of contents

1. Introduction
2. Objectives
3. Company/Organization Profile
4. Project developed
5. Skills and Knowledge Gained
6. Challenges
7. Conclusion

Introduction

I. Purpose of report

This report summarizes my one-month internship experience at Machdatum as an Embedded Developer intern from 7th May to 7th June 2025. The internship aimed to apply academic knowledge in a real-world setting, while learning new skills in communication, protocols, microcontrollers, and data management. The focus was on understanding processes and products in the field of Industry 4.0..

II. Overview of the internship

In consideration of the guidelines provided by the university, the company I chose to work with is a startup I interned at Machdatum, a startup based in Coimbatore, Tamil Nadu, founded in 2018. The company specializes in Industry 4.0 solutions for manufacturing and process industries. I worked in the IoT department, where I was responsible for designing a project to collect, store, and manipulate data from industrial machinery using microcontrollers and tools like MQTT, React, and PostgreSQL. This experience helped bridge the gap between academic concepts and industry practices in embedded systems and data management.

III. Company Background

Machdatum Ltd. is a software development startup founded in 2018, located in Coimbatore, Tamil Nadu. It focuses on solutions for asset management, work order management, and inventory tracking. Its products include *ThingManage* (a low-code IoT data pipeline) and *ThingSight* (a business intelligence tool for shop-floor performance).

Objectives

I. Program Outcomes

The university's 19MEE490 Summer Internship program aimed to help students apply engineering knowledge to solve complex problems, propose solutions using modern tools, and consider ethics and sustainability. The program also focused on developing skills in communication, teamwork, project management, and finance..

II. Internship Outcomes

As an intern, I worked on a project involving microcontrollers and embedded systems to tackle a real-world problem. The main goals were to design an innovative solution, understand project requirements, and gain proficiency in necessary tools and technologies. Key objectives included:

- Designing and researching solutions for the problem
- Building hardware and coding with Arduino
- Transferring data to the cloud using MQTT protocol
- Manipulating data with PostgreSQL
- Applying machine learning models for predictions
- Using computer vision techniques to improve image/video quality
- Learning web frameworks to provide a user interface service

III. Project outcome

- The project focused on home security automation to detect theft, intruders, and fires. Key outcomes included:
- Designing a network security system for both interior and exterior monitoring
- Collecting and transferring data in real-time with minimal errors
- Detecting intruders, unauthorized entry, and fire hazards using computer vision and machine learning, while minimizing false alarms
- Developing a user-friendly website to log and display data

Company profile

Machdatum Ltd., founded in 2018 and located in Coimbatore, Tamil Nadu, is a privately held software development startup. The company focuses on providing Industry 4.0 solutions to improve the efficiency of manufacturing and process industries using technologies like IoT, Data Engineering, Machine Learning, and AI. Their clients include MNCs such as ZF Wind Power and Shanthi Gears.

Machdatum's products include *ThingManage*, a low-code IoT data pipeline enabling seamless connectivity between field devices and software systems, and *ThingSight*, a business intelligence tool for shop-floor management. They also provide a CMMS system for work order management, asset management, and real-time inventory tracking.

The company organized a one-day session to introduce their products and workflows. During my internship in the IoT department, I worked with the embedded development team to implement IoT solutions. The team was led by experts such as Dr. Ramasamy M (Managing Director), Mr. Hemanand Ramasamy (CEO), and other team members like Noufal Basheer (IoT Lead) and Kavya P (Embedded Engineer), who guided me throughout the internship by providing learning materials and assignments to deepen my understanding of the project.

Description of Internship Duties

I. Weekly responsibilities

Week	Work
1	Familiarize with microcontroller basics, Learn and work on the Internal ADC Design a project pipeline for a problem with a solution based on IoT Intro to Industrial IoT 4.0 Understand the working of the products developed by the company
2	Draft the requirements (software and hardware) for the projects Work with data acquisition with Arduino and development modules Familiarize with Communication protocols Work with MQTT to push the data to the cloud Draw the network system of the house
3	Learn PostgreSQL for data manipulation Set the data to time series data using TimescaleDB. Python code to push data to SQL
4	Train a machine learning and computer vision integrating model and test it using the data collected from Arduino in SQL Learn node.js and JavaScript Create a simple dashboard to retrieve data and the prediction of output from sql and ML node, respectively, to the user Draft a paper on the project topic

II. Project pipeline

The project focuses on automating a full-fledged home security system, covering both the interior and exterior of the house.

Exterior Security:

CCTV cameras are installed at key points around the house. Computer vision techniques are used for automation:

- **Object Detection:** Moving objects are detected using Non-Max Suppression, and HOG (Histogram of Oriented Gradients) descriptors are used for human detection.
- **Face Recognition:** Captured faces are checked against known family member images using OpenCV's Haar Cascade and DeepFace for face verification. If a recognized family member is detected, no alerts are triggered. If an intruder is detected, motion is tracked using `cv2.createBackgroundSubtractorMOG2`, which detects changes by subtracting the background.

Interior Security:

Non-invasive sensors (temperature, light, sound, CO2, and PIR) are used to detect intruders. These sensors are adjusted based on the house size:

- **Intruder Detection:** A machine learning model, based on a PyTorch deep learning neural network, processes the sensor data. If the system detects a sudden increase in occupancy or abnormal sensor activity, alerts are sent to the homeowner.
- **Anomaly Detection:** Isolation Forest is used to detect abnormal sensor readings, such as changes indicating a house fire.

System Integration:

Both the exterior and interior systems are integrated to form a comprehensive home security system.

The combined solution provides automated alerts and real-time monitoring for any suspicious activity.

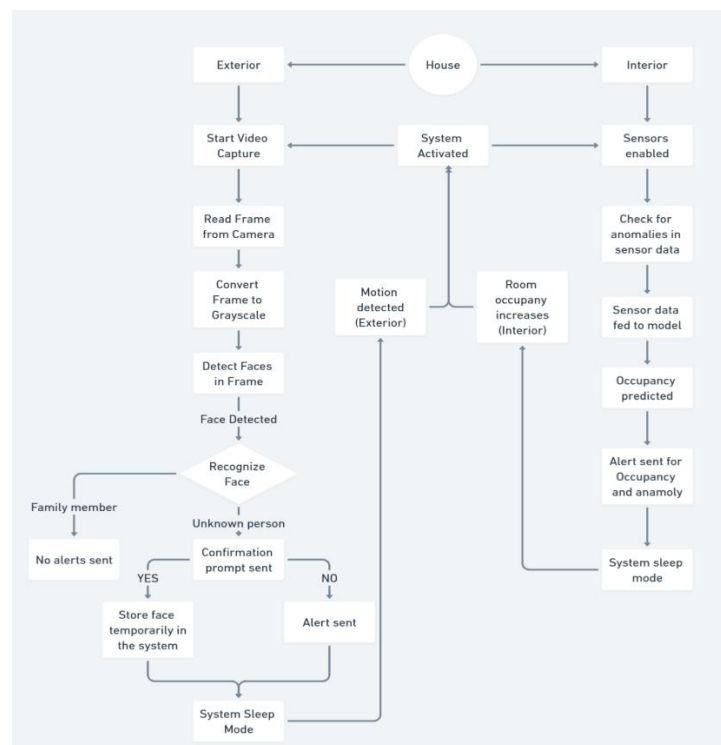


Fig 1. Logic of the automated security system

III. Hardware connections

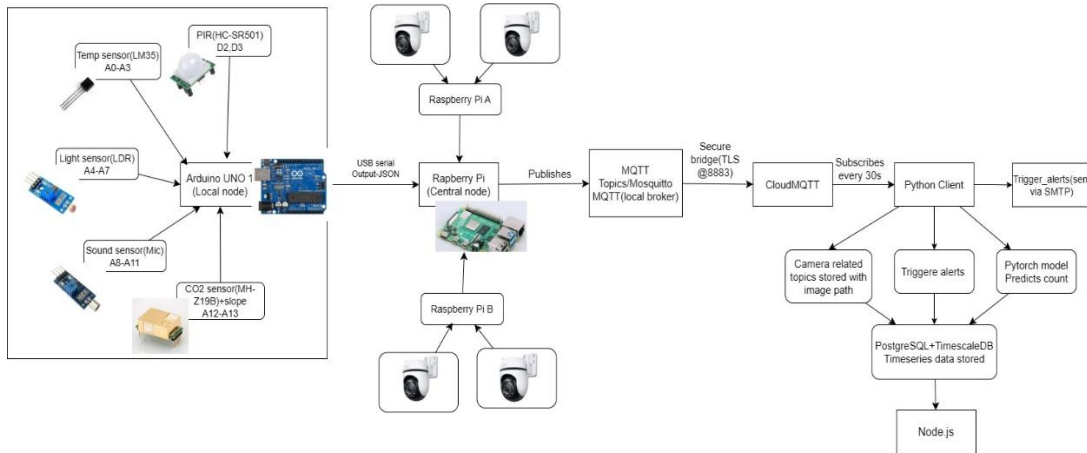


Fig.2 Backend of system

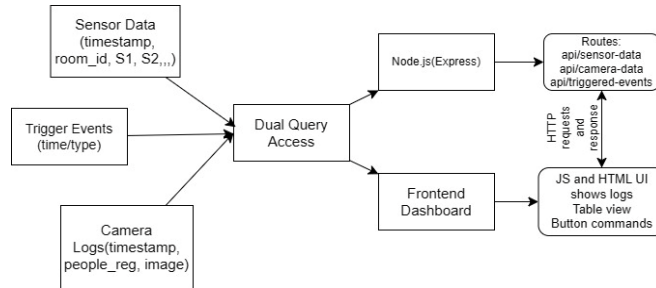


Fig.3 Frontend of system

A) Sensor → local Arduino:

Each room with a sensor suite was connected to a common Arduino node to collect data, a typical room of size 6m x 4.6m utilizes a combination of Sensors which includes S1_Temp, S2_Temp, S3_Temp, S4_Temp, S1_Light, S2_Light, S3_Light, S4_Light, S1_Sound, S2_Sound, S3_Sound, S4_Sound, S5_CO2, S5_CO2_Slope, S6_PIR, S7_PIR.

Table 1 : Type of sensor and its connection with the arduino

Sensor	Type	Quantity	Pins
Temperature	Analog (LM35)	4	A0-A3
Light	Analog (LDR + resistor), Voltage divider	4	A4-A7
Sound	Analog (mic sensor), Analog output to ADC	4	A8-A11
CO ₂ + Slope	Analog (MH-Z19B)	1	A12, A13
PIR	Digital (HC-SR501)	2	D2, D3

With the change of room size, a certain number of sensors can be reduced or increased.

B) Raspberry Pi Communication(Central node):

The Raspberry is considered the central node, combining the exterior and interior segment.

The Arduino is coded to print sensor data in JSON format, Raspberry listens to the USB serial port, it reads JSON, parses it, and prepares it for further processing. The option for choosing GPIO-to-GPIO UART was not chosen because of the voltage mismatches, raspberry uses 3.3V logic and Arduino UNO uses 5V logic a voltage divider or level shifter must be used, so the alternate was chosen which was USB

On the other hand, the CCTV footages are captured. If any motion is detected in the cameras with (cv2.createBackgroundSubtractorMOG2), the footages are immediately sent for review to the central node.

Step 1: Capture CCTV feeds, using IP cameras

Step 2: Motion detection (MOG2) (Raspberry Pi A and B)

Step 3: Object/Human Detection (HOG + NMS): OpenCV's cv2.HOGDescriptor() + SVM people detector, apply Non-Maximum Suppression to refine detection (Raspberry Pi A and B)

Step 4: Face Detection (cv2.CascadeClassifier) (Raspberry Pi A and B)

Step 5: Face Recognition (DeepFace.verify) (Central Raspberry node)

Step 6: The prediction pushed to cloud.

C) Transferring messages with MQTT:

The final sensor data and the Raspberry Pi camera module data are stored in the central node. The MQTT runs with the logic of subscription and publishing of messages to topics

Table 2: Topics in MQTT

Data Type	Topics	Format
Sensor Data (Room 1)	home/room1/sensors	JSON
Sensor Data (Room 2)	home/room2/sensors	JSON
Motion Detected	home/security/motion/raspA/camera1	Boolean/String
Human Detected	Home/security/human/raspA/camera1	Boolean/String
Face Verified	home/security/face/verified	JSON (with ID, status)
Face Unknown	home/security/face/unverified	JSON + cropped image path/url

The data from the central node is pushed to the server via the MQTT local broker Mosquitto to bridge it to CloudMQTT, so any data that is published locally can be published in the cloud. Encryption is added via the command “bridge_cafile /etc/ssl/certs/ca-certificates.crt” and is run in the port 8883.

D) Using and storing data from CloudMQTT → ML model and PostgreSQL

Python client in the model subscribes to CloudMQTT topics, incoming JSON data every 30 sec is parsed and is passed to the pretrained PyTorch-based deep learning neural network model as a test file with the same number input features utilized for training for prediction. The prediction (room_occupancy_count) is combined with the sensor data and stored in PostgreSQL. The timescaleDB creates a time series data in PostgreSQL, that is the sensor data with the time series and prediction is stored in tables. Continuous aggregate view feature is to summarize the data over time in chunks, and a retention policy is set up to store only one month's data.

E) Creating a simple dashboard to retrieve the data

A simple dashboard using Node.js was built with JavaScript and HTML. To retrieve the data requested by the user.

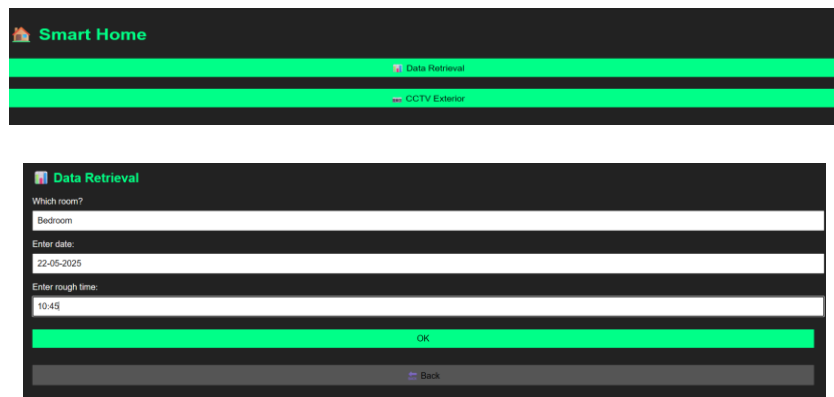


Fig 4. Web dashboard

IV. Tools Used

Hardware:

- 1) 3x Raspberry Pi
- 2) 1x Arduino uno
- 3) 4x LM35
- 4) 2x HC-SR501
- 5) 1x MH-Z19B
- 6) 4x mic sensor

7) 4x LDR + resistor

Software:

- 1) Arduino IDE
- 2) ThingSpeak
- 3) MQTT explorer
- 4) Timescaledb PostgreSQL
- 5) PyCharm/VSCoDe
- 6) Pytorch
- 7) Javascript
- 8) HTML and CSS

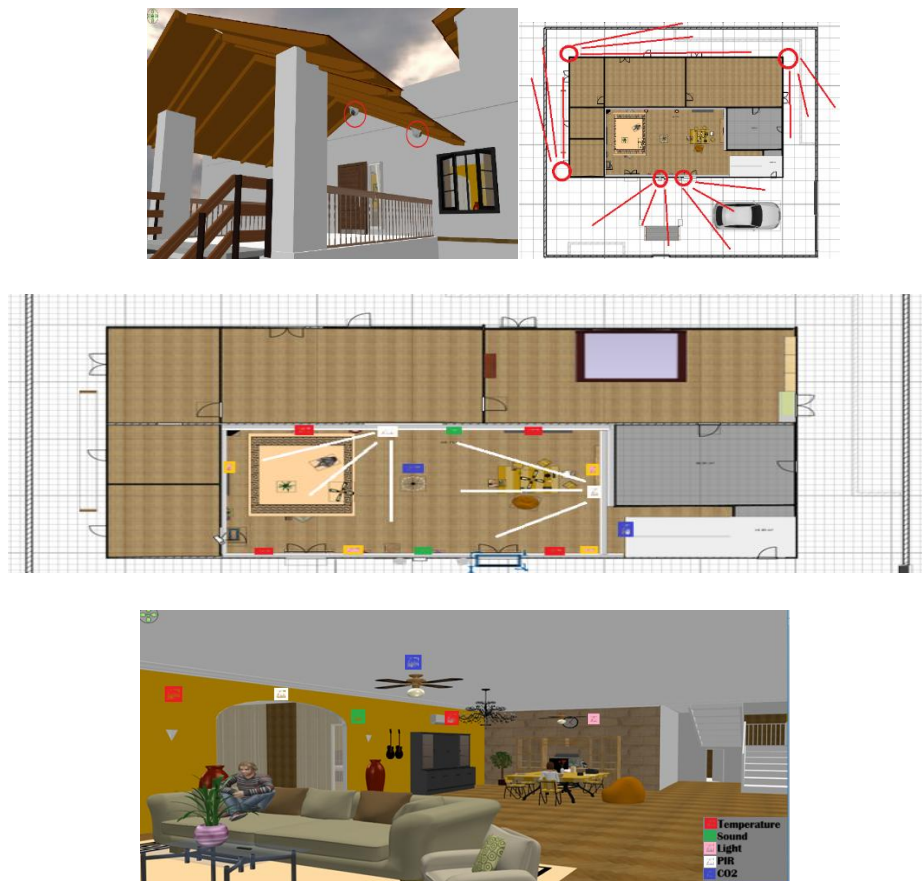


Fig.5 CCTV and Sensor placement

Skills and Knowledge Gained

Skills acquired or improved during the summer internship are of,

Technical:

- Working with Arduino and development modules
- Data transfer to the cloud using the MQTT protocol
- Manipulating data with PostgreSQL
- Utilizing data to identify predictions with Machine learning
- Utilizing computer vision and filtering techniques to improve image or video quality for analysis.
- Web frameworks to provide service to customers using JavaScript and Node.js

Interpersonal:

- Awareness
- Collaborating and working
- Communication skills
- Socializing skills

Organizational:

- Designing a project and organizing workflows
- Understanding the objectives for the project and planning the learning to implementation cycle

Challenges Faced

During my internship, I encountered several challenges, both technical and task-related. One of the main difficulties was debugging errors in the code. As a beginner, it was tough to identify the issues, especially since the entire project relied heavily on coding, ranging from Arduino programming to web development. These errors often slowed my progress and made it hard to meet deadlines. However, I gradually learned effective error-solving techniques and discovered that using logging functions significantly streamlined the debugging process and saved a lot of time.

Communication was another challenge for me. As an introvert, I initially found it difficult to express myself or ask for help. Fortunately, the team was very welcoming and supportive, which made it easier for me to open up and communicate more comfortably over time.

Adapting to the work environment was also a bit difficult at first. Sitting in one place for 8 hours a day was quite new to me, but I found that taking regular walk breaks helped me stay refreshed and focused.

Lastly, since I was new to many of the tools and software used in the project, I had to dedicate extra time to learning and familiarizing myself with them. Despite the initial struggle, consistent research and hands-on learning enabled me to successfully contribute to and complete the project.

Conclusion

Overall, it was a new and fulfilling experience; It helped me implement the knowledge I gathered from college with a project, which helped me understand the concepts from a different perspective. I was introduced to a variety of new tools and technologies, which have significantly influenced my career interests and direction. Through the internship, I also developed better planning and organizational skills, which helped me recognize my capabilities and potential. Additionally, experiencing a professional work environment gave me valuable insight into workplace culture, communication, and collaboration.